TD8 : Convolution

Ex 1:

1. Calculer $\chi_{[0,1]} * \chi_{[0,1]}$ $\in L^1(\mathbb{R})$

soit $x \in \mathbb{R}$,

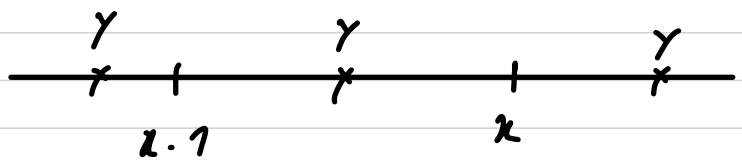
$$\chi_{[0,1]} * \chi_{[0,1]}(x) = \int_{\mathbb{R}} \chi_{[0,1]}(x-y)\, \chi_{[0,1]}(y)\, dy$$

$$= \int_0^1 \chi_{[0,1]}(x-y)\, dy$$

$\chi_{[0,1]}(x-y) = 1$

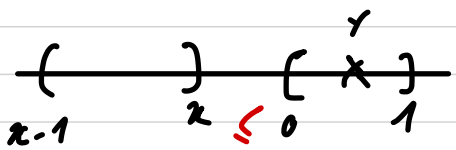
$\Leftrightarrow \quad 0 \leq x - y \leq 1$

$\Leftrightarrow \quad -1 + x \leq y \leq x \qquad \Leftrightarrow \quad y \in (x-1, x] \cap [0,1)$



$y \in [0,1)$

i)

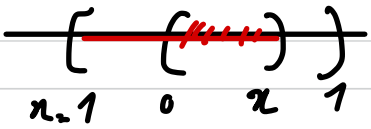


$\Rightarrow \chi_{[0,1]}(x-y) = 0$

$\Leftrightarrow \quad x \leq 0$

$\Rightarrow \chi_{(0,1)} * \chi_{(0,1)}(x) = 0$

ii)

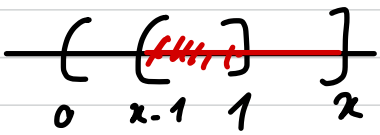


$\Rightarrow \chi_{(0,1)} * \chi_{(0,1)}(x) = x$

$\Leftrightarrow \quad x \in [0,1)$

$$\int_0^x 1\,dy = x$$

iii)



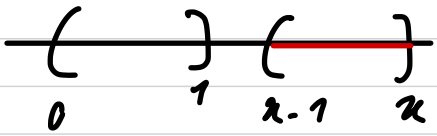
$$\chi_{(0,1)} * \chi_{(0,1)}(x) = \int_{x-1}^{1} 1\,dy$$

$$= 1-(x-1)$$

$$= 2-x$$

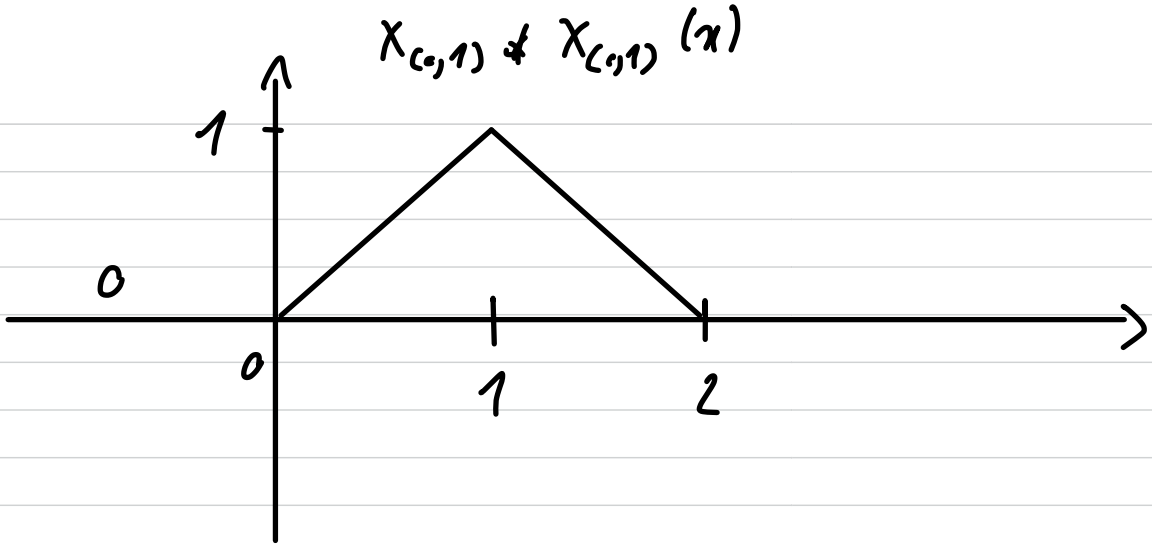
$\Leftrightarrow \quad 1 \leq x \leq 2$

iv)



$$\chi_{(0,1)} * \chi_{(0,1)}(x) = 0$$

$\Leftrightarrow \quad x \geq 2$



ex: $\left(\chi_{(0,1)} * \chi_{(0,1)}\right) * \chi_{(0,1)}$ ($*$ associatif)

<u>Autre Méthode :</u>

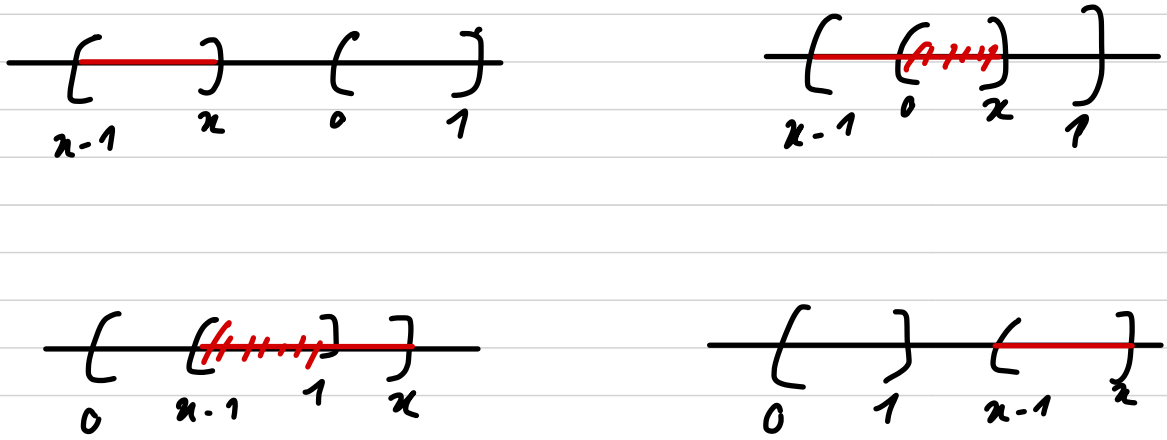
$$\chi_{(0,1)} * \chi_{(0,1)}(x) = \int_0^1 \chi_{(0,1)}(x-y)\,dy$$

$$z = x-y \in (x-1, x]$$

$$\begin{cases} y = \varphi(z) = x - z \\ \varphi'(z) = -1 \\ |\det \varphi'(z)| = |-1| = 1 \end{cases}$$

$$\Rightarrow \int_0^1 \chi_{(0,1)}(x-y)\,dy$$

$$= \int_{(x-1,x)} \chi_{(0,1)}(z)\,|-1|\,dz$$



2. $\begin{cases} \chi_{(-a,a)} * \cos \\ \chi_{(-a,a)} * \sin \end{cases}$ $(a>0)$

Calculons pour cela $\chi_{(-a,a)} * e^{ix}$

Soit $x \in \mathbb{R}$,

$$\underbrace{\chi_{(-a,a)}}_{L^1(\mathbb{R})} * \underbrace{e^{ix}}_{\in L^\infty(\mathbb{R})} = \int_{\mathbb{R}} \chi_{(-a,a)}(x-y)\, e^{iy}\, dy$$

$\in L^1(\mathbb{R})$ $\left(\int_{\mathbb{R}} \overbrace{|e^{ix}|}^{1}\, dx = \infty\right)$

(cf $|e^{ix}| = 1$)

* commutatif

$$= e^{ix} * \chi_{(-a,a)}$$

$$= \int_{\mathbb{R}} e^{i(x-y)} \chi_{(-a,a)}(y)\, dy$$

$$= \int_{-a}^{a} e^{i(x-y)}\, dy$$

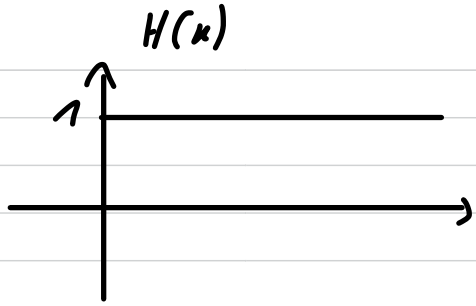
$$= \left[-\frac{1}{i} e^{i(x-y)} \right]_{-a}^{a}$$

$$= \frac{e^{ix}\left(e^{ia} - e^{-ia}\right)}{i}$$

$$= 2\sin(a)\, e^{ix}$$

$$\Rightarrow \begin{cases} \chi_{(-a,a)} * \cos = 2\cos(x)\sin(a) \\ \chi_{(-a,a)} * \sin = 2\sin(x)\sin(a) \end{cases}$$

3. $H = \chi_{\mathbb{R}_+} \in L^\infty(\mathbb{R})$



Heaviside

si $f \in L^1(\mathbb{R})$, $f * H(x) = \int_{\mathbb{R}} f(x-y)\, H(y)\, dy$

$$= \int_{\mathbb{R}_+} f(x-y)\, dy$$

$z = x - y \in]-\infty, x]$

$y = \varphi(z) = x - z$

$|\det \varphi'(z)| = |-1| = 1$

$$= \int_{]-\infty; x]} f(z)\, |-1|\, dz$$

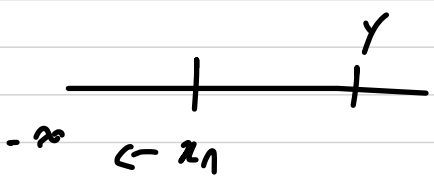
$$= \int_{-x}^{x} f(z)\, dz$$

On obtient la primitive de f qui a pour limite 0 en $-\infty$; en effet, soit $(x_n)_n \to -\infty$,

$$H * f(x_n) = \int_{-\infty}^{x_n} f(y)\,dy$$

$$= \int_{\mathbb{R}} \underbrace{f \cdot \chi_{]-\infty ; x_n]}(y)}_{f_n(y)}\,dy$$

Soit $y \in \mathbb{R}$ fixé, $f_n(y) \xrightarrow[n \to \infty]{} 0$



i.e. $(f_n)_n$ CVS vers 0

$$|f_n(y)| \leq |f(y)| \in L^1(\mathbb{R})$$

CV dominée $\Rightarrow$ $\lim \int f_n = \int \lim f_n = 0$

Exo 2 :

1. Soient $f \in L^1(\mathbb{R})$ et $g \in L^\infty(\mathbb{R})$;

mq $f * g(x) = \int_{\mathbb{R}} f(x-y)\, g(y)\, dy$

bien définie, $\in L^\infty(\mathbb{R})$ et que

$$\|f * g\|_\infty \leq \|f\|_1 \cdot \|g\|_\infty$$

soit $x \in \mathbb{R}$ fixé,

$$\int_{\mathbb{R}} |f(x-y)\, g(y)|\, dy = \int_{\mathbb{R}} |f(x-y)|\, |g(y)|\, dy$$

$$\leq \underbrace{\|g\|_\infty}_{g \in L^\infty(\mathbb{R})} \int_{\mathbb{R}} |f(x-y)|\, dy$$

on pose $z = x - y$

$$\begin{cases} y = \varphi(z) = x - z \\ \varphi'(z) = -1 \\ |\det \varphi'(z)| = |-1| = 1 \end{cases}$$

$$\int_{\mathbb{R}} |f(x-y)\, g(y)|\, dy \leq \|g\|_\infty \int_{\mathbb{R}} |f(z)|\, dz$$

on $f \in L^1(\mathbb{R})$

donc $\int_{\mathbb{R}} |f(z)|\, dz < \infty$

donc $\int_{\mathbb{R}} |f(x-y)\, g(y)|\, dy < \infty$

ainsi $f * g(x)$ bien définie

de plus, soit $x \in \mathbb{R}$,

$$|f * g(x)| = \left| \int_{\mathbb{R}} f(x-y)\, g(y)\, dy \right|$$

$$\leq \int_{\mathbb{R}} |f(x,y)\, g(y)|\, dy$$

$$\leq \|f\|_1 \, \|g\|_\infty$$

$\Rightarrow f * g \in L^\infty(\mathbb{R})$ et $\|f*g\|_\infty \leq \|f\|_1 \cdot \|g\|_\infty$

2. Soient f et $g \in L^2(\mathbb{R})$, mq $f*g$ bien définie

et que $f*g \in L^\infty(\mathbb{R})$ avec $\|f*g\|_\infty \leq \|f\|_2 \cdot \|g\|_2$

soit $x \in \mathbb{R}$,

$$\int_{\mathbb{R}} |f(x-y)\, g(y)|\, dy < \infty$$

On, $\int_{\mathbb{R}} |f(x-y)||g(y)|\, dy$

$$\underset{\text{Hölder}}{\leq} \left(\int_{\mathbb{R}} |f(x-y)|^2 dy\right)^{\frac{1}{2}} \cdot \underbrace{\left(\int_{\mathbb{R}} |g(y)|^2 dy\right)^{\frac{1}{2}}}_{\|g\|_2}$$

Comme $\int_{\mathbb{R}} |f(x-y)|^2 dy \underset{z=x-y}{=} \int_{\mathbb{R}} |f(z)|^2 dz = \|f\|_2^2$

$$\Rightarrow \int_{\mathbb{R}} |f(x-y)\, g(y)| \leq \underbrace{\|f\|_2 \, \|g\|_2}_{\text{indépendant de } x} < \infty$$

bien définie

$$\Rightarrow |f * g(x)| = \left|\int_{\mathbb{R}} f(x-y)\, g(y)\, dy\right|$$

$$\leq \int_{\mathbb{R}} |f(x-y)\, g(y)|\, dy$$

$$\leq \underbrace{\|f\|_2}_{\text{indépendant de } x} \|g\|_2$$

$\Rightarrow f * g \in L^\infty(\mathbb{R})$

(+) $\|f*g\|_\infty \leq \|f\|_2 \|g\|_2$

Exo 3 :

1. $f \cdot g \in L^1(\mathbb{R})$

mq $\widehat{f*g} = \hat{f} \cdot \hat{g}$

On a, pour $\xi \in \mathbb{R}$,

$$\widehat{f*g}(\xi) = \int_{\mathbb{R}} e^{-2i\pi\xi x} f*g(x)\,dx$$

$$= \int_{\mathbb{R}} e^{-2i\pi\xi x} \left(\int_{\mathbb{R}} f(x-y)\, g(y)\, dy \right) dx$$

posons $\varphi(x,y) = e^{-2i\pi\xi x} f(x-y)\, g(y)$

si $\varphi \in L^1(\mathbb{R}^2)$, Fubini s'applique et on aura :

$$\int_{\mathbb{R}^2} \varphi(x,y)\, dx\, dy$$

$$= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \varphi(x,y)\, dy \right) dx$$

$$= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{-2i\pi\xi x} f(x-y)\, g(y)\, dy \right) dx$$

$$\left(= \widehat{f * g}(\xi) \right)$$

$$= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \varphi(x,y)\, dx \right) dy$$

$$= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \underbrace{e^{-2i\pi \xi x}}_{e^{-2i\pi \xi (x-y)} \quad e^{-2i\pi \xi y}} f(x-y) \, g(y) \, dx \right) dy$$

$$= \int_{\mathbb{R}} e^{-2i\pi \xi y} g(y) \left(\int_{\mathbb{R}} e^{-2i\pi \xi \overbrace{(x-y)}^{z = x-y}} f(x-y) \, dx \right) dy$$

$$= \int_{\mathbb{R}} e^{-2i\pi \xi y} g(y) \, \hat{f}(\xi) \, dy$$

$$= \hat{f}(\xi) \underbrace{\int_{\mathbb{R}} e^{-2i\pi \xi y} g(y) \, dy}_{\hat{g}(\xi)}$$

Vérifions donc que $\varphi \in L^1(\mathbb{R}^2)$

$$\int_{\mathbb{R}^2} |\varphi(x,y)| \, dx \, dy < \infty$$

$$= \int_{\mathbb{R}^2} |\overset{1}{\cancel{e^{-2i\pi\xi x}}} f(x-y) \, g(y)| \, dx \, dy$$

$$= \int_{\mathbb{R}^2} |\underbrace{f(x-y) \, g(y)}_{\geq 0}| \, dx \, dy$$

$$\overset{\nearrow}{=} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x-y)| \, |g(y)| \, dx \right) dy$$

Fubini ≥ 0
(Tonelli)

$$= \int_{\mathbb{R}} |g(y)| \left(\int_{\mathbb{R}} |f(\underbrace{x-y}_{z = x-y})| \, dx \right) dy$$

$$\rightarrow \int_{\mathbb{R}} |f(z)| \, dz = \|f\|_1$$

$$= \|f\|_1 \underbrace{\int_{\mathbb{R}} |g(y)|\, dy}_{\|g\|_1} < \infty$$

2. $\chi_{[0,1]} * \chi_{[0,1]} * \chi_{[0,1]}$

$$= \left(\widehat{\chi_{[0,1]}}(\xi) \right)^3$$

cf TD 7